The Hierarchical Bayesian Growth Model

Implementation, and comparison with the hidden Markov model

Parv

Supervisor: Prof. K. R. Sahasranand

Department of Data Science, IIT Palakkad

August 2026

Abstract

The hierarchical Bayesian growth model specified in the proposal (pages 13–15) has been implemented as written and compared against the hidden Markov model from Weeks 5–7. Neither replaces the other; they answer different questions, and the comparison is informative in both directions.

On predicting the next recorded subtype, the hidden Markov model is ahead in **6 of 6** independent splits, by $+0.058 \pm 0.019$ log-likelihood per transition. The reason is specific and measurable: the growth model's residual standard deviation is $\sigma = 1.138$, which is **4.6 times the width of the entire Indeterminate band**, so band membership is dominated by noise.

The growth model nevertheless supplies something no Markov model can: an interpretable progression rate on the clinical scale. The TD/PIGD ratio declines by **0.093** per year (95% credible interval 0.068 to 0.116), with a between-patient standard deviation of 0.103 per year — the individualised trajectory the proposal is ultimately after.

Two problems with the model as specified were found by implementing it. The observation and process variances are provably not separable, and a Normal model on the raw ratio places 12.2% of its five-year predictive mass on negative ratios, which are impossible.

1 The model, as specified

From the proposal, pages 13–15:

$$R_{it}^{obs} = R_{it} + \epsilon_{it}^{obs}, \qquad \epsilon_{it}^{obs} \sim \mathcal{N}(0, \sigma_{obs}^2) \tag{1}$$

$$R_{it} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})\, t + \epsilon_{it}^{proc}, \qquad \epsilon_{it}^{proc} \sim \mathcal{N}(0, \sigma_{proc}^2) \tag{2}$$

$$b_{0i} \sim \mathcal{N}(0, \tau_0^2), \qquad b_{1i} \sim \mathcal{N}(0, \tau_1^2) \tag{3}$$

with the clinical translation

$$P(\text{TD}) = P(R > 1.15), \quad P(\text{Ind}) = P(0.90 \le R \le 1.15), \quad P(\text{PIGD}) = P(R < 0.90).$$

Inference is by Gibbs sampling with conjugate updates throughout (`src/ppmi_subtype/hbayes.py`); no external probabilistic-programming dependency. A full run over 573 patients takes about one second. The sampler is validated by simulating trajectories from chosen parameters and checking the posterior covers them (`tests/test_hbayes.py`).

1.1 A note on the two variance terms

Substituting (2) into (1):

$$R_{it}^{obs} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})\, t + \underbrace{(\epsilon_{it}^{proc} + \epsilon_{it}^{obs})}_{\text{variance } \sigma_{proc}^2 + \sigma_{obs}^2}$$

Both error terms are independent and additive on the *same* observation, so the likelihood depends on them only through their sum. Holding $\sigma^2_{proc}+\sigma^2_{obs}$ fixed and moving mass between the two leaves the log-likelihood numerically unchanged — verified directly. They are **not separately identifiable** in this parameterisation.

The sampler therefore estimates the identified quantity $\sigma^2 = \sigma^2_{proc} + \sigma^2_{obs}$ rather than reporting a split the data cannot support. This matters for the headline output: $P(TD) = P(R^{true} > 1.15)$ refers to the *latent* ratio, whose spread excludes observation noise. Without separating the two, only the predictive distribution of R^{obs} is identified.

Separating them needs either the state-space form the proposal also mentions (where process noise accumulates over time and so leaves a distinguishable signature) or repeated measurements at a single visit.

2 Data

Labelled visits		4,441
With a finite TD/PIGD ratio	3,716	(83.7%)
Without (PIGD subscore is zero)		725
Training observations / patients		2,426 / 573

Table 1: The growth model is defined on the ratio, so the 725 visits with a zero PIGD subscore cannot enter it. They are 665 TD and 60 Indeterminate, so the loss is not random — it removes the most tremor-dominant visits.

The Markov models of Weeks 5–7 have no such restriction, which is why the comparison in Section 4 is confined to transitions both models can score.

3 Fitted model

Parameter	**Mean**	**SD**	**2.5%**	**97.5%**
β_0 population baseline ratio	2.2920	0.0669	2.1577	2.4272
β_1 population progression, /yr	−0.0927	0.0121	−0.1159	−0.0682
σ total residual SD	1.1380	0.0203	1.0993	1.1778
τ_0 SD of patient baselines	1.3408	0.0499	1.2466	1.4421
τ_1 SD of patient slopes, /yr	0.1033	0.0137	0.0780	0.1315

Table 2: Posterior summary, 2,000 retained draws. First and second halves of the chain agree to four decimals on every parameter.

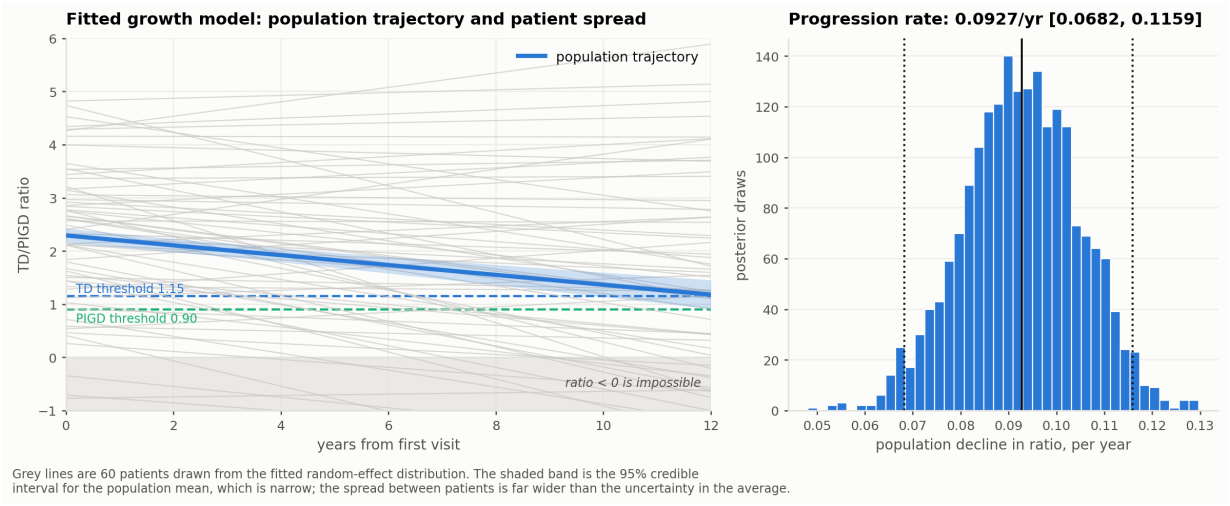


Figure 1: The fitted model. Left: the population trajectory with its credible interval, and 60 patients drawn from the random-effect distribution. Right: the posterior for the population progression rate.

The population trajectory declines, at 0.0927 per year with $P(\beta_1 < 0) = 1.000$. Starting from a mean baseline ratio of 2.292, the population average reaches the TD threshold after a median **12.3 years** (95% CI 9.9 to 16.4) and the PIGD threshold after 15.0 years (12.1 to 19.9).

Patients differ substantially. The between-patient SD of the slope is 0.103 per year, comparable in size to the population slope itself. A patient one standard deviation faster than average progresses at roughly −0.196 per year, reaching the TD threshold in about six years rather than twelve.

4 Head-to-head with the hidden Markov model

Both models are fitted on the same 755 training patients and scored on the same held-out transitions — those where every visit involved has a finite ratio.

Model	Log-lik / transition	Accuracy	Brier
HMM (latent states)	**−0.6559**	**0.761**	**0.3573**
Hierarchical Bayesian growth	−0.6814	0.737	0.3786

Table 3: 825 shared held-out transitions.

Split	n	HMM	HB growth	Difference
0	761	−0.6948	−0.7350	+0.0403
1	832	−0.6822	−0.7544	+0.0722
2	817	−0.7111	−0.7897	+0.0785
3	778	−0.7264	−0.7731	+0.0467
4	748	−0.7028	−0.7760	+0.0732
5	786	−0.7355	−0.7717	+0.0362

Table 4: Six independent patient-level splits, both models refitted each time. The HMM is ahead in 6 of 6, mean +0.0579 (sd 0.0188). Paired t-test $p = 6.6 \times 10^{-4}$; Wilcoxon $p = 0.031$.

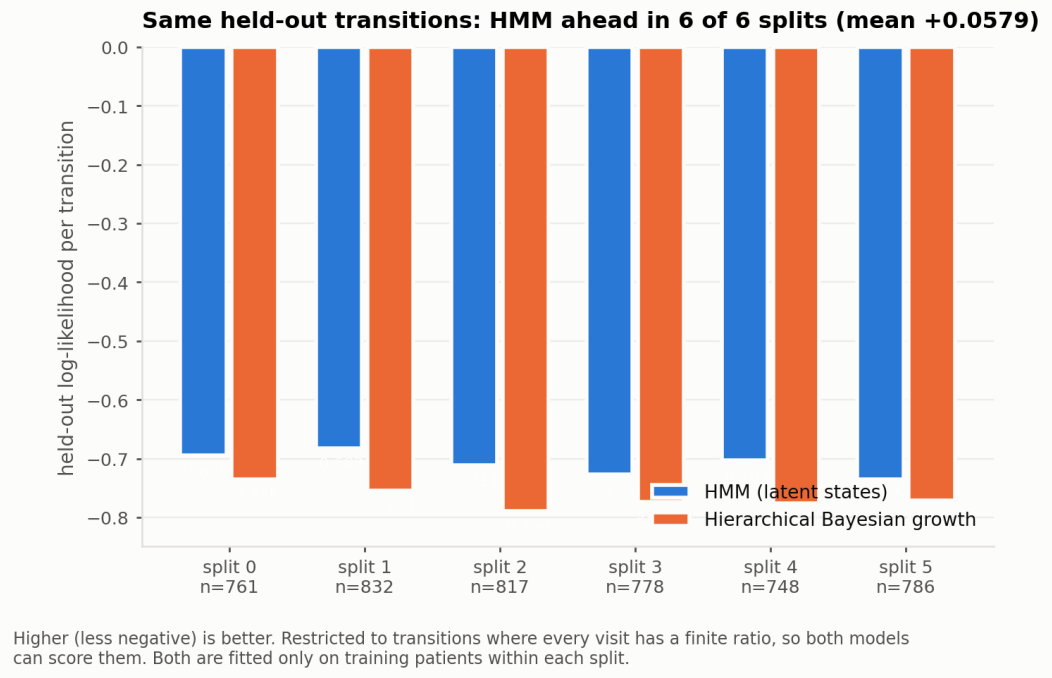


Figure 2: The same comparison. Higher is better.

4.1 Why the growth model loses at this task

The total residual SD is $\sigma = 1.138$. The Indeterminate band is $1.15 - 0.90 = 0.25$ wide. **The noise is 4.6 times the width of the band the model must predict membership of.** Even with a perfectly estimated trajectory, the next observation lands almost anywhere, so the Indeterminate class is nearly unhittable. The Markov model does not face this: it predicts the class directly and never has to locate a continuous value inside a narrow interval.

A second symptom of the same mismatch: a Normal observation model on the raw ratio places **12.2% of its five-year predictive mass below zero**, and a ratio of two non-negative scores cannot be negative. The ratio is also strongly right-skewed (skew 1.27, range 0 to 11.36), which a symmetric Normal cannot represent.

4.2 The two clinical tables side by side

	HMM (latent states)			HB growth		
Horizon	TD	Ind	PIGD	TD	Ind	PIGD
1 year	0.910	0.071	0.019	0.759	0.052	0.189
2 years	0.834	0.111	0.055	0.741	0.052	0.206
3 years	0.769	0.133	0.099	0.714	0.061	0.224
5 years	0.660	0.149	0.191	0.680	0.052	0.268

Table 5: For a patient currently TD (HMM) or at the population baseline ratio of 2.29 (growth model).

The growth model gives a patient whose ratio is 2.29 — twice the TD threshold — an 18.9% chance of being PIGD within one year. That is not clinically credible, and it is the σ problem showing up directly in the deliverable. Note also that its probabilities barely move with the horizon, because the noise dominates the trend over five years.

5 What the growth model provides that the Markov model cannot

The comparison above is only about one task. On everything else the growth model is the more useful object.

Quantity	HMM	HB growth
Predicts the next recorded subtype well	**yes**	weaker
Progression rate on the clinical scale, with CI	no	**yes**
Patient-specific progression rate	no	**yes**
Estimated years to cross a threshold	no	**yes**
Uses visits with an undefined ratio (16.3%)	**yes**	no
Quantifies label misclassification	**yes**	no
Extends to the σ_i stability phenotype	no	**yes**
Accepts baseline covariates naturally	awkward	**yes**

The Markov model reports *that* a patient is likely to change state. The growth model reports *how fast their phenotype is moving*, on the clinical scale, with uncertainty — and lets that rate differ between patients. The second half of the proposal, in which patient-specific variability σ_i becomes a phenotype in its own right, is only expressible in the growth model.

6 Recommendation

1. **Keep both.** They are not competing answers to one question. The Markov model is the better classifier of the next visit; the growth model is the better description of disease progression.
2. **Model the ratio on a transformed scale.** The three symptoms above — σ far exceeding the band width, negative predictions, strong right skew — all follow from putting a symmetric Normal on a non-negative, skewed quantity. A log or similar transform would address all three at once, and the thresholds carry over exactly, since $R > 1.15 \iff \log R > \log 1.15$. This needs care because the ratio is exactly zero at some visits, so a shifted transform or a censored treatment is required.
3. **Resolve the two-variance question deliberately.** Either adopt the state-space form, where the two noise sources are distinguishable, or state in the manuscript that only the total is identified. The current specification cannot support reporting them separately.
4. **Model the two subscores rather than their ratio.** This would recover the 16.3% of visits that have no finite ratio, and would remove the zero-denominator problem at its source.
5. **Then add covariates and σ_i.** Once the observation scale is settled, the second half of the proposal follows directly.

Reproducing this report.

```
python scripts/03_progression_model.py     # HMM and Markov baselines
python scripts/04_hierarchical_bayes.py    # growth model and comparison
python -m pytest tests/ -q                 # 134 tests
```

Stage 4 writes to `outputs/hbayes/`. Stage 3 was not modified.